

Paper 17.75 - Error-Correction Tower Next-State Bridge

CQE-paper-17.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-17.75.md

Paper 17.75 - Error-Correction Tower Next-State Bridge

Purpose

This bridge tells later papers how to use Paper 17 as a preconditioned state. The exported object is a verified tower with open glue and extractor residues, not a completed rank-24 construction.

Exported State

Paper 17 exports:

```
verified chain: 1 -> 3 -> 7 -> 8 -> 24 -> 72
verified Hamming/Fano rung
verified extended Hamming/E8 seed
verified Golay ingredient carrier
verified powered Nebe-72 K-bound
verified E8^3 root-shell landing
open Leech glue action
open W(E8) extractor
open semantic terminal map
```

Legal Use In Paper 18

Paper 18 may use Paper 17 to frame representation and glue routes if it keeps the following rule:

```
verified rung + explicit residue -> next-state candidate
```

That means the `24` rung can seed a Leech discussion only as Golay ingredient geometry until glue action is actually constructed.

Illegal Use

Do not write:

```
Paper 17 proves the Leech lattice.
Paper 17 proves W(E8) extraction.
Paper 17 proves physical fault correction.
Paper 17 maps every N to a terminal lattice.
```

Those statements erase the boundary receipts and are therefore invalid.

Recenter Rule

When a later paper uses the tower, recenter at the rung actually being used:

```
C = selected rung
B = boundary condition at that rung
T = allowed transition to the next closure frame
O = residue that did not close
```

The next paper may then repeat the read, but it must not inherit a stronger claim than this bridge exports.

Hidden-Guess Diagnostic

Before revealing a route, ask the reviewer or local agent to predict whether the route is:

```
closed
closed-with-residue
ingredient-only
open
invalid
```

Reveal the receipt after the prediction and record the mismatch as training data.

Conclusion

Paper 17.75 carries the tower forward as a disciplined next-state object. It lets Paper 18 build from the verified rungs while keeping Leech, Weyl, and terminal-selection obligations visible.